

LOVE SALMON

2024

SUSTAINABILITY REPORT

Canadian Aquaculture Industry Alliance

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OVERVIEW

OUR FIRST SUSTAINABILITY REPORT



THIS IS THE INAUGURAL SUSTAINABILITY REPORT FOR CAIA'S SALMON FARMERS, WHICH IS CENTERED AROUND SEVEN NATIONAL, INDUSTRY-WIDE COMMITMENTS FOR SALMON FARMING, AND THEIR ASSOCIATED ACTIONS AND TARGETS.

These commitments outline priority areas for continuous improvement in our operations. The data in this report covers the period from January 1, 2022, to December 31, 2022.

In line with our seventh national salmon farming commitment (i.e., Public Reporting), we intend to publish this report annually going forward and ultimately be able to provide multi-year performance data for all actions and targets. Case studies are included throughout the report to highlight successes and progress towards meeting our commitments and demonstrating how Canada's salmon farmers are global leaders in the sustainable production of farm-raised salmon.

The scope of this report includes data from five[1] of CAIA's salmon-producing members and their Canadian operations (herein referred to as "CAIA salmon farmers" when discussing data points). Operations include freshwater land-based hatcheries and ocean farms, and is limited to primary processing. Ocean farms are the most common grow-out method of salmon aquaculture in Canada, accounting for approximately 99.5% of Canada's farm-raised salmon production.

For clarity, all currency, unless otherwise stated, is reported in Canadian dollars ($CAD).

The development of this report is governed by the CAIA Competition Policy. No discussions took place that included any exchange of any current or future supply, price or market information.

CAIA acknowledges the important contributions of salmon farmers, industry experts, partners, and stakeholders who were critical to the development of our sustainability commitments and this report.

We welcome your feedback on our sustainability report and initiatives. Please contact info@aquaculture.ca with any questions. You may also visit aquaculture.ca and lovesalmon.ca for more information about CAIA and our sustainability commitments.

[1] Two additional companies are anticipated to participate in the next report. However, it was not feasible this year given various reporting constraints.

MESSAGE FROM THE PRESIDENT & CEO

I AM VERY HAPPY TO INTRODUCE THIS FIRST ANNUAL REPORT FOR THE CANADIAN SALMON FARMING SECTOR'S NATIONAL PERFORMANCE COMMITMENTS.

This is the product of many months of work and collaboration across the sector, in order to demonstrate to Canadians our commitment to transparency and constant improvement. Ocean-farmed salmon make up about 99.5% of all the farmed salmon produced in Canada, and about 95% of the total salmon (wild and farmed) harvested in Canada. Canadians and the world want more Canadian-raised salmon, but they need to know it is of the highest quality and sustainability. Each company has its own sustainability systems and reporting, and all salmon farms are independently certified to global or Canadian standards. This report is an important additional tool to help build this knowledge across Canada.

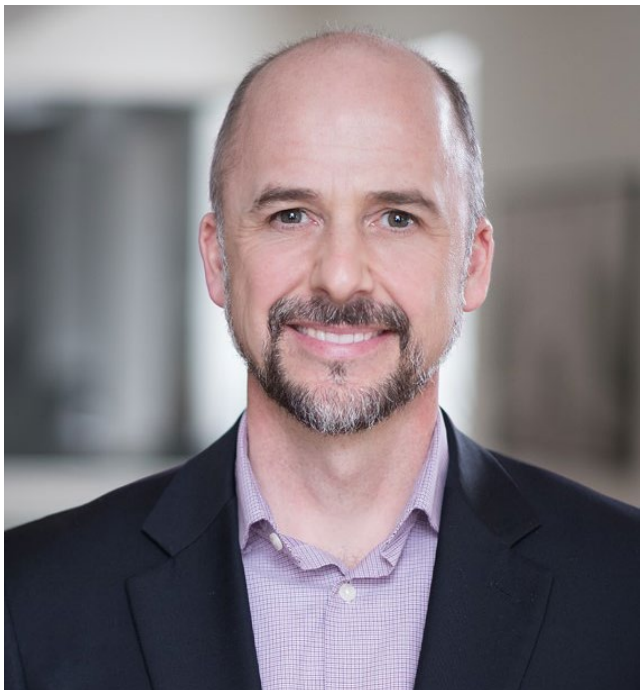
As we work on improving and communicating performance, we also celebrate successes. This report outlines some of these. For example, escapes from farms in 2022 were at a historic low, the carbon footprint of farmed salmon is the lowest of all large-scale farmed animal species, and our product is 100% traceable from egg to plate.

Our goal is to be nothing less than the most sustainable animal product in the world, show why and how, and continually work towards further improvement. While we have made great progress in ensuring our operations perform at the highest level, there is always more to do. For example, we want to further reduce fish mortality; we want all of our fish feed to be 100% from sustainable-certified sources; we want to reduce our handling of fish to reduce stress and improve fish health; we want to further expand our work to support strong and vital wild salmon populations.

The good news is that with a predictable regulatory system and clear objectives, we will continue to invest in innovation and new technologies that will further advance the performance of the sector. There are many existing technologies that can be adopted into our sector very quickly that support meeting the challenges above. We have committed to reporting against our objectives every year. This is the first report, and with each year we plan and hope to see improvements and progress in all areas. Climate change and warming waters are a major challenge for all of us and some of these impacts are unpredictable, but we know that with a strong investment environment, we can respond to challenges and provide a healthy, secure, and high-quality Canadian source of food for all Canadians.

For more information on our commitments, actions, and targets, visit the Love Salmon website. **(www.lovesalmon.ca)**

TIMOTHY J. KENNEDY
Canadian Aquaculture Industry Alliance



Funding for this project has been provided through the AgriAssurance Program under the Canadian Agricultural Partnership, a federal-provincial-territorial initiative.

Visit **www.lovesalmon.ca** to follow our story.

ABOUT THE

CANADIAN AQUACULTURE INDUSTRY ALLIANCE

CAIA is the national association that speaks for Canada's seafood farmers, representing their interests in Ottawa to regulators, policymakers, and political leaders. With a membership that reaches coast to coast to coast, comprised of finfish, shellfish, and aquatic plant farmers, feed companies and suppliers, as well as regional aquaculture associations, CAIA is a passionate advocate for the quality and sustainability of farm-raised seafood.

CAIA's salmon farmers operate 108 farms across five provinces, contributing to local economies, and providing local employment opportunities in rural, coastal, and Indigenous communities. Collectively, CAIA's reporting members directly employed 3,984 full-time equivalent employees in Canadian operations, and many thousands more in indirect/induced jobs on Canada's East and West coasts.

In 2021, CAIA's salmon-producing members began discussions to develop a set of national, industry-wide commitments. In 2022, CAIA established an External Advisory Panel (EAP) to provide advice and feedback on the commitments, actions, and targets as they were being developed. The EAP informed the industry's collective commitments to sustainability practices, performance as well as disclosure and communications.

CAIA MEMBERS' SALMON FARMING OPERATIONS

Five CAIA members participated in this report. All CAIA members are expected to participate in the next report.

Cermaq Canada
BC - *Atlantic Salmon*

Grieg Seafood BC Ltd.
BC - *Atlantic Salmon*

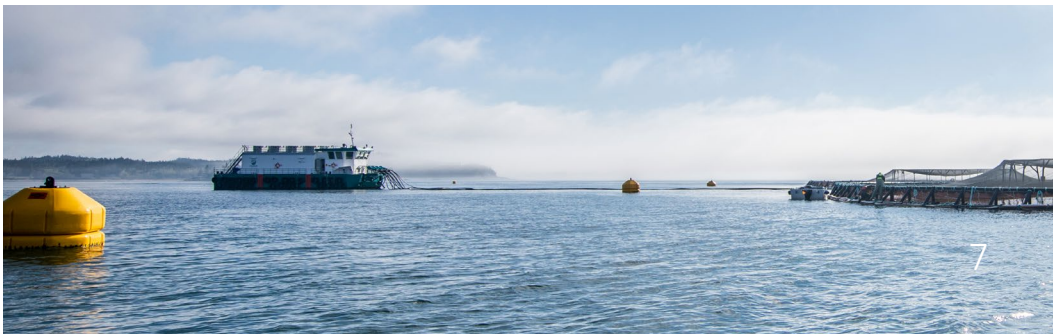
Mowi Canada West
BC - *Atlantic Salmon*

Cooke Aquaculture Inc
NB, NL, & NS - *Atlantic Salmon*

Grieg NL Seafarms Ltd.
NL - *Atlantic Salmon*

Mowi Canada East
NB & NL - *Atlantic Salmon*

Creative Salmon Co. Ltd.
BC - *Chinook Salmon*



OUR PERFORMANCE





HIGHLIGHTS

As our inaugural Canadian sustainability report, this document marks the beginning of our sustainability disclosure journey. Below are some of the key results for 2022. Our approach to sustainability is centred around seven national, industry-wide commitments for salmon farming, and their associated actions and targets, which touch on the themes of fish health, climate change, sustainable feed, food security, food traceability, ocean health, and public reporting. For more information on our commitments, actions, and targets, visit the **Love Salmon website (www.lovesalmon.ca).**

Our sustainability disclosure journey is a work in progress, and we want to acknowledge and be transparent that companies are still working to identify the right indicators for all commitment areas, and, for some indicators, there is missing company data. Companies are working to implement data collection and management practices to be able to report on all commitment areas and indicators in future years.

100% Compliance with the Code of Practice for the Care and Handling of Farmed Salmonids

81.88% Fish survival rate in the ocean

0.065kg Antibiotic use - kilograms (kg) of active ingredient per tonne of live weight fish produced

34 Instances where farms were required to take mitigative action to reduce sea lice levels*

13 Number of farm-raised fish escape incidents in the marine environment

(110 fish escaped total)

70% Percent of all sea lice treatments being non-medicinal

5 Number of companies working with partners to minimize interactions between farm-raised and wild fish

5 Number of companies providing donations or in-kind support for wild salmon recovery or enhancement programs

0.58 Greenhouse gas (GHG) intensity - metric tonnes of (tCO_2e) generated per tonne of harvest weight fish produced**

5 Number of companies implementing management practices to adapt to and mitigate the impacts of climate change

** 4 companies reporting, ** 3 companies reporting,*



92.14%

Percentage of fish meal and fish oil in feed coming from certified sustainably managed compliant fisheries



87.95%

Percentage of feed derived from non-marine nutritional ingredients



49.15%

Percentage of non-fish protein in feed that is certified according to sustainability programs*



96.13%

Utilization rate of the salmon



153K

Number of salmon meals donated in Canada**



$1,2M+

Donations to non-profit organizations and First Nations that support food security amongst Canadians*



99.3%

Ocean farms equipped with underwater feed system monitoring equipment



100%

Percentage of farm sites remediated before a new generation is transferred in



4

Number of companies that have made efforts to pursue restorative aquaculture, multi-species aquaculture, and/or co-production opportunities



86.47%

Percentage of packaging used in primary processing that is reusable, recyclable, or biodegradable**

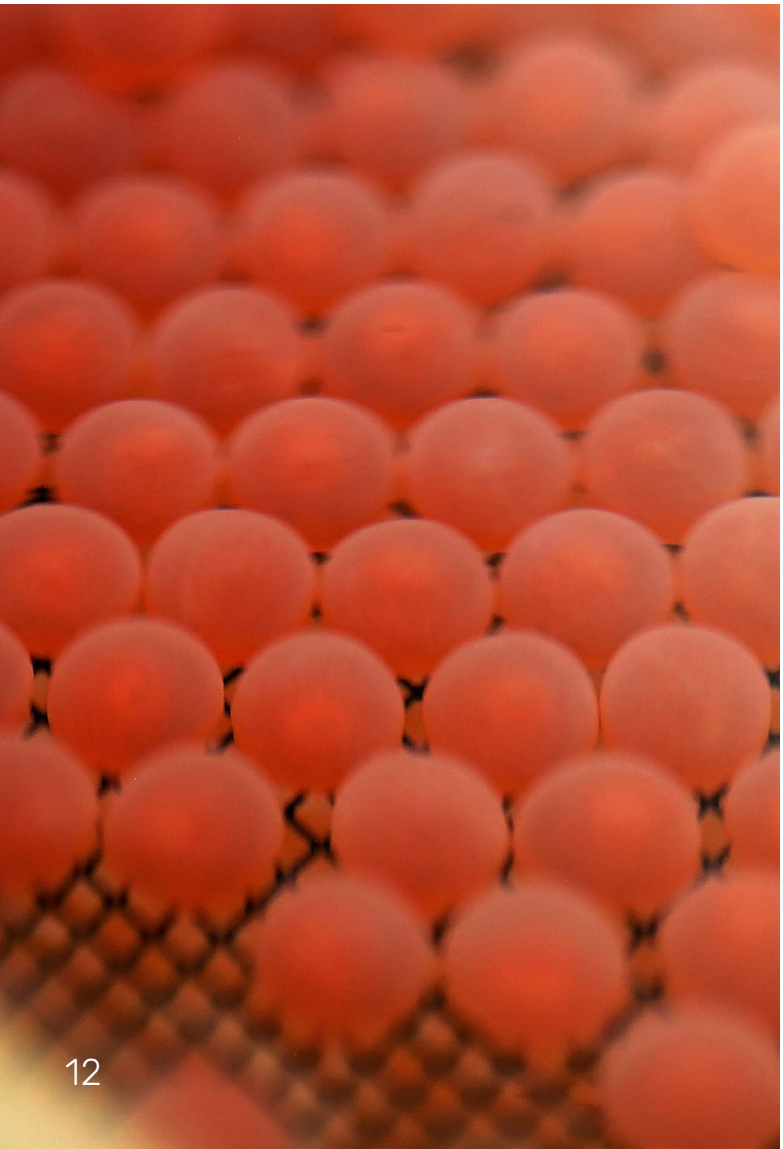
** 4 companies reporting, ** 3 companies reporting*



FISH HEALTH

COMMITMENT

The health and welfare of farm-raised and wild salmon is a top priority. We are committed to using best management practices and technologies that support the well-being of farm-raised salmon. We will also pursue a variety of partnerships, measures, and initiatives that support wild salmon recovery, habitat restoration, and enhancement efforts. Performance against relevant actions and targets are summarized below.



EMBRACE CANADA'S CODE OF PRACTICE FOR THE CARE AND HANDLING OF FARMED SALMONIDS

The Code of Practice for the Care and Handling of Farmed Salmonids (the "Code") was released in 2021. As the first fish welfare code in Canadian history, it provides an important benchmark for animal health and welfare indicators.

OUR PERFORMANCE

Animal welfare standards and practices are incorporated throughout operations, as evidenced through corporate animal welfare policies, operations protocols and standards, fish health management plans, employee training, and documentation practices. 100% of CAIA salmon farmers are in compliance with the Code.

TARGET 90% FARM-RAISED FISH SURVIVAL IN OCEAN FARMS

In the wild, fish survival rates are challenged by natural predators, disease, and environmental factors, all of which contribute to overall low survival rates. On the farms, where fish health and safety are the cornerstone of successful operations, survival rates are much higher.

OUR PERFORMANCE

The fish survival rate measures the number of fish that survived over the year as a proportion of stocked fish. In 2022, CAIA salmon farmers achieved an 81.88% survival rate in ocean farms, accounting for fish mortalities and culls. Fish survival rates on farms are typically impacted by algal blooms, abnormal weather patterns, low oxygen events, and predator attacks. In comparison, the annual average survival rate for farm-raised salmon in the three largest production regions (Norway, Chile, Scotland) was 81% in 2022.

CONTINUED REDUCTION IN ANTIBIOTIC USE

The use of vaccines and improved health management practices have enabled a significant reduction in the use of antibiotics in salmon farming over the past decades. To prevent disease, salmon are reared in an environment that limits stress and reduces susceptibility to pathogens and pests.

OUR PERFORMANCE

CAIA salmon farmers cumulatively used 0.065 kg of active pharmaceutical ingredients per metric tonne of live weight fish. Used as a last resort, salmon antibiotics treat against outbreaks of naturally occurring, disease-causing bacteria. These treatments can only be applied when approved by Health Canada and administered by a veterinarian. We have worked to implement science-based best management practices, such as using appropriate stocking densities and proactively vaccinating fish, to reduce the need of antibiotics across farms. Canadian salmon farming is a low user of antibiotics. More information on Canadian livestock utilization rates can be found **here (Canada.ca)**.

COMPLIANCE WITH SEA LICE REGULATIONS AND PROTOCOLS, INCLUDING INTEGRATED PEST MANAGEMENT (IPM) STRATEGIES

Sea lice are a naturally occurring ocean parasite that feed on the skin and mucous of both wild and farm-raised salmon. All salmon farmers in Canada have pest management plans in place, employing a variety of preventative and treatment measures.

OUR PERFORMANCE

CAIA salmon farmers must routinely report on sea lice at active salmon farms. As part of compliance, farms are required to take mitigative action to reduce sea lice levels if found to be greater than the regulatory threshold in their jurisdiction. In 2022, there were 34 instances where farms were required to take mitigative action to reduce sea lice levels.[2] Non-medicinal treatments are the primary treatment method, with 70.87% of all sea lice treatments being non-medicinal, and medicinal treatments are only used when necessary. Many of these non-medicinal treatments use the biology of both the sea lice and salmon to control lice populations with minimal environmental impact (e.g., cleaner fish or freshwater baths).

CASE STUDY

Taking a lesson from nature, CAIA salmon farmers provide their salmon freshwater baths at sea. The small skin parasite known generally as 'sea lice' dies when exposed to freshwater – a natural lifecycle of the parasite/host relationship when fish move from sea to river systems.

Technological developments now enable salmon farmers to replicate this made-in-nature solution to sea lice exposure that also improves gill health by producing pure freshwater (reverse osmosis) contained in large custom-built vessels.

Pressure pumps gently draw salmon into the vessel where they are immersed in clean freshwater supplemented with oxygen for a few hours. The tiny sea lice are killed, drop off the salmon, and captured with microfilters for proper disposal. At the same time, the fish's gills (lungs for fish) are flushed and cleaned.



[2] *4 companies reporting (of 5 companies).*

ACHIEVE ZERO ESCAPES ANNUALLY

Minimizing the interactions that farm-raised salmon have with the surrounding environment is one of the ways to avoid potential interactions with wild salmon and the ecosystem. Rare failures in farm infrastructure have resulted in salmon escaping into the ocean. In Canada, the majority of salmon farmers must report suspected and confirmed fish escapes directly to regulators.

OUR PERFORMANCE

In 2022, CAIA salmon farmers experienced 13 escape incidents, resulting in a confirmed total of 110 escaped fish in the marine environment. Causes of escape incidents included human error during sea lice sampling and harvesting, predator attacks (i.e., seals), and mechanical or equipment failures, such as having holes in a catch-net during mechanical delousing.

CASE STUDY

As seal populations grow, CAIA salmon farmers have experienced an increase in attempts by seals to enter ocean farms. This population growth causes greater competition for naturally occurring food resources, and so seals make bolder attempts to breach nets, increasing the risk of fish escapes. Beyond the proper installation of main netting and an additional layer of netting to dissuade predators, producers have installed stainless steel core netting at many sites where seal sightings are frequent and populations are abundant. This additional measure deters predator intrusion by preventing seals from chewing through the netting material. When the seal realizes they cannot breach the net, they move on to seek other food sources.

WORK WITH FIRST NATIONS, GOVERNMENTS, AND OTHER PARTNERS TO DEEPEN UNDERSTANDING OF MARINE ENVIRONMENTS AND IMPLEMENT NEW TECHNOLOGIES AND INNOVATIVE APPROACHES TO ANIMAL HUSBANDRY THAT MINIMIZE FARM-RAISED AND WILD FISH INTERACTIONS AND MAINTAIN ECOSYSTEM HEALTH

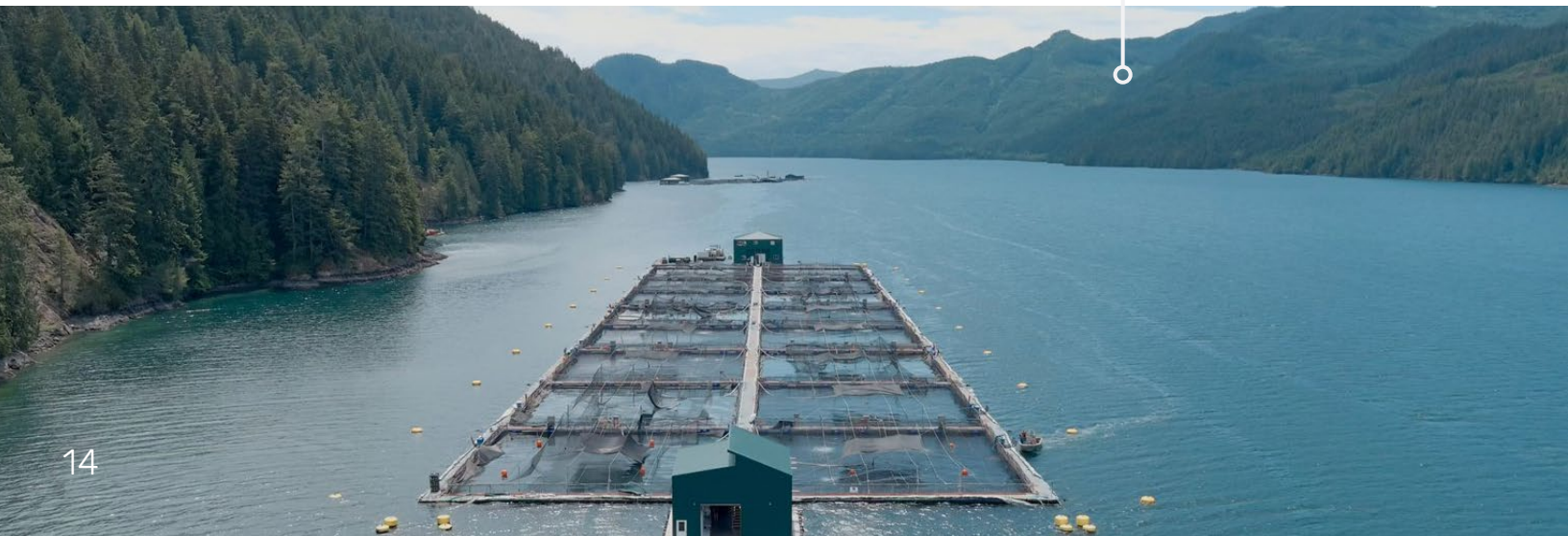
While numerous scientific reviews show that the impact of farms on wild salmon populations is low, a better understanding of marine ecosystem dynamics will contribute to solutions that further reduce disease and pest transmission, flowing from either wild to farm-raised or farm-raised to wild salmon.

OUR PERFORMANCE

To minimize interactions between farm-raised and wild fish and maintain ecosystem health, CAIA salmon farmers are researching, developing, and testing new separation technologies. These include sea lice skirts, bubble curtains for plankton, and new containment systems.

CASE STUDY

Grieg Seafood is trialling a made-in-BC semi-closed technology solution at its farms. They have completed installation of semi-closed systems at three of its farms, off the west coast of Vancouver Island. The new CO2L Flow system (pronounced Cool Flow) is a form of semi-closed containment, which allows farmers to raise or lower custom-designed farm enclosures – ensuring the farmed fish benefit from natural ocean conditions while also providing protection for wild salmon. While in this trial phase, farmers expect better growth, lower mortality, better feed conversion rates (meaning the fish are more effective at converting feed into growth), and a dramatic reduction in the need for sea lice treatments.



SUPPORT LOCAL AND INDIGENOUS WILD SALMON RECOVERY, HABITAT RESTORATION AND ENHANCEMENT INITIATIVES THROUGH RESEARCH, FUNDING, AND IN-KIND SUPPORT

Over many decades, wild salmon stocks have been in decline. Salmon farming was, in part, a response to this decline in wild populations. There are numerous actions that can meaningfully contribute to the support of wild salmon, many of which benefit from the contribution of expertise and experience that are shared by salmon farmers.

OUR PERFORMANCE

Through donations or in-kind support, CAIA's salmon farmers actively support wild salmon recovery or enhancement programs. This includes wild fish sampling for sea lice, and financial support and equipment donations to local salmon enhancement organizations.

CASE STUDY

Creative Salmon is supporting a local First Nation with a Hatchery-in-a-Box, contributing to the purchase of the unit and consultant assistance on the project. A Hatchery-in-a-Box is a transportable, compact, mini-hatchery and so less expensive than a traditional, permanent hatchery. The hatchery can be set up quickly, be placed in remote locations, and requires less staffing. It is hoped that use of the Hatchery-in-a-Box for wild salmon enhancement will contribute to more robust local wild salmon populations.





CASE STUDY

A revolutionary wild salmon partnership – Fundy Salmon Recovery – includes Fort Folly First Nation, Parks Canada, Cooke Aquaculture, Atlantic Canada Fish Farmers Association, the University of New Brunswick, and the Province of New Brunswick. Together, they have pioneered the world's first wild salmon marine conservation farm on Grand Manan Island, New Brunswick. This is where wild salmon, collected from the inner Bay of Fundy rivers as juveniles, are grown to maturity and then released back into their native rivers. The goal is that these fish will naturally spawn the next generation of wild salmon, providing their offspring with early exposure to the wild and therefore the best chance of survival as adults. The partnership work has led to numbers of naturally returning and spawning Atlantic salmon in participating rivers not seen in decades. While in the conservation farm, these salmon are cared for by staff from Cooke Aquaculture. Fundy Salmon Recovery is the first project in the world to collect juvenile fish after they have spent the critical early life stages in the wild, then transport those fish to be cared for in an ocean environment at a dedicated marine farm site. The fish are fed specialized diets developed by biologists and aquaculture nutrition experts.



CLIMATE CHANGE

COMMITMENT

Farm-raised salmon has the lowest carbon footprint of any large-scale farmed animal protein, and we know we can do more. We are committed to continuing to reduce our carbon emissions and are equally committed to becoming more resilient in the midst of a changing environment. Performance against relevant actions and targets are summarized below.

SUPPORT CANADA'S COMMITMENT TO NET-ZERO BY ACHIEVING A 35% REDUCTION IN GHGS BY 2030

Despite farm-raised salmon's low carbon footprint, we are targeting further reductions in greenhouse gas (GHG) emissions through technology advancement in areas such as renewable energy, the incorporation of electric equipment, and sustainable shore-based power solutions.

OUR PERFORMANCE

In 2022, CAIA salmon farmers measured direct GHG emissions (also known as Scope 1 emissions, as classified by the GHG Protocol Corporate Standard) occurring from sources that are owned and controlled by companies. This includes emissions from:

- *The generation of electricity, heat, or steam as part of farm operations (land and ocean based),*
- *Transportation of materials, products, waste, and employees within the farm operations boundary, or where the company has an exclusive lease with a contractor and 100% of transportation is related to their operations, and*
- *Fugitive emissions, whether intentional or unintentional, such as emissions from equipment leaks, hydrofluorocarbon releases during the use of refrigeration, and air conditioning equipment.*[3]

Salmon farmers emitted a total of 42,075.40 metric tonnes of direct carbon dioxide equivalent (tCO_2e) generated directly through farm operations.[4] The GHG intensity of farm-raised salmon in Canada is 0.58 tCO_2e per tonne of harvest weight fish produced.

According to the Global Salmon Initiative, farm-raised salmon has the lowest carbon footprint, measuring GHG emissions per typical serving of edible protein, when compared to other large-scale farmed animal protein sources, such as chicken, pork, or beef.

Salmon farmers are investing in the research, trial, and deployment of low-carbon technologies and energy-saving initiatives to further reduce the carbon footprint of their operations.

CARBON FOOTPRINT OF ANIMAL PROTEIN SOURCES

Carbon footprint measures the total GHG emissions caused by the production of a product, measured by the Global Salmon Initiative as kilograms of carbon dioxide equivalent per typical serving of edible protein of the product ($kgCO_2e/40g$).

Source: Global Salmon Initiative, 2022



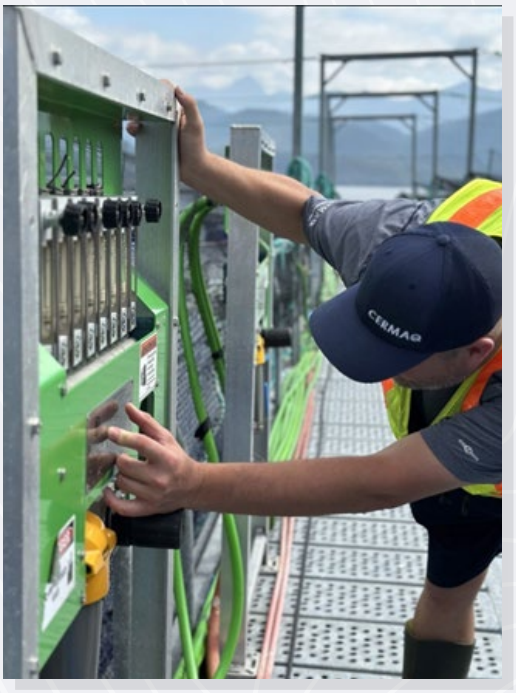
[3] *Direct GHG emissions from processing operations or product transportation to the customer are not included within the scope.*
[4] *3 companies reporting (of 5 companies).*

CASE STUDY

New aeration technology helped Cermaq Canada cut over 76 tCO_2e emissions from their operations in 2021. The reduction in emissions was the result of trialing Canadian innovator Poseidon Ocean System's new Flowpressor™ aeration compressor systems at two of Cermaq's 16 operational farm sites in British Columbia. The trial took place at the Simmonds Point and Wehlis Bay farms from May through November 2021, and involved using Flowpressors to power Cermaq's standard Harmful Algae Bloom systems instead of conventional diesel compressors.

"The trial demonstrated the viability of the Flowpressor technology to power our existing conventional algae bloom mitigation equipment, while producing 60% less greenhouse gas emissions. The overall result is that we have slashed emissions and dramatically reduced our fuel costs while improving overall performance of the mitigation system," said Brock Thomson, Cermaq Canada's Innovation Director.

Due to the success of the trial, Cermaq Canada now has Flowpressor systems installed at all of its Canadian sites.



CONTINUE TO DEVELOP AND ADOPT TECHNOLOGIES AND PRACTICES THAT BUILD THE CAPACITY OF PRODUCERS TO ADAPT TO A CHANGING CLIMATE

Warming waters, increased storm and weather event intensity and changing salinities all contribute to changes in the ecology and abundance of many resident species surrounding salmon farms. Marine aquaculture operations are well positioned to increase monitoring of the surrounding environments where farms are located and thus contribute to a better understanding and more effective mitigation efforts in dealing with changes.

OUR PERFORMANCE

In 2022, CAIA salmon farmers implemented management practices to adapt to and mitigate the impacts of climate change. Examples of these management practices included:

- *Integrating lice skirts to help prevent the spread of sea lice, which proliferate in warmer waters as a result of climate change,*
- *Conducting real-time environmental data logging to better understand mitigation and adaptation efforts, and*
- *Completing a Task Force on Climate-Related Financial Disclosures risk assessment to better understand climate-related risks that are material to salmon farming.*





SUSTAINABLE FEED

COMMITMENT

We support innovation in aquaculture feeds and are committed to the sourcing of environmentally sustainable ingredients that provide high-quality nutrition for our fish. Performance against relevant actions and targets are summarized below.

WORK WITH OUR FEED PRODUCERS TO ENSURE 100% OF THE FISH MEAL AND FISH OIL IN OUR FEED COME FROM CERTIFIED SUSTAINABLY MANAGED/FAO-COMPLIANT FISHERIES

Fish meal and fish oil are important nutritional ingredients in most commercial fish feeds, providing a balanced amount of essential amino acids, phospholipids, and fatty acids. Ensuring that the fish meal and fish oil is being harvested and processed in a sustainable way is critical in ensuring the sustainability of the sector.

OUR PERFORMANCE

92.14% of fish meal and fish oil in feed come from certified sustainably managed and/or United Nations Food and Agriculture Organization (FAO)-compliant fisheries. Sustainably managed fisheries are certified according to standards such as MarinTrust and the Marine Stewardship Council, which state that marine raw materials shall not originate from illegal, unregulated, and unreported catch or from fish species classified as endangered by the International Union for the Conservation of Nature red list. FAO-compliant fisheries are in compliance with the FAO Code of Conduct for Responsible Fisheries.



SUPPORT THE ACCELERATED DEVELOPMENT AND ADOPTION OF SUSTAINABLE ALTERNATIVE INGREDIENT SOURCES THAT OPTIMIZE FISH HEALTH, PRIORITIZING CANADIAN SUPPLIERS WHERE POSSIBLE

While fish meal is the most natural diet for farm-raised salmon, significant research and development is going into non-marine protein alternatives. Given the pressure on certain wild fish stocks traditionally used for fish meal, finding viable alternatives that maintain a well-balanced diet and continue to produce a nutritious final product will contribute to the overall sustainability of the sector.

OUR PERFORMANCE

87.95% of feed comes from non-marine nutritional ingredients. To support a reduction in the use of marine feed ingredients, CAIA salmon farmers continue to look for new non-marine nutritional ingredients that could be used in feed. Companies have conducted independent trials of novel ingredients such as algal oil, camelina, canola oil, and others. Sustainability of non-fish protein is also important. 49.15% of non-fish protein is certified according to sustainability programs (e.g., Roundtable for Responsible Soy, ProTerra, Sustainable Palm Oil, or equivalent).[5]

CASE STUDY

All of CAIA's salmon farmers recognize the importance of making aquaculture feed less dependent on marine nutritional ingredients. Under its certification from the Canadian Organic Aquaculture Standard, Creative Salmon's Chinook diet must follow specific parameters, but the standard does make allowances for some feed ingredient substitutions. With that in mind, the company has undertaken feed trials on site at their ocean farms. Inclusion of black soldier fly larvae meal, algal, canola, and camelina oil in the feed is showing some promise that novel non-marine nutritional ingredients alternatives may become more routine in the future. More research is required.

[5] *4 companies reporting (of 5 companies).*



FOOD SECURITY

COMMITMENT

Farm-raised salmon have a key role in helping to feed Canada and the world as capture fisheries reach their limits. Global population growth and demand for seafood continue to accelerate creating pressure on wild stocks of all finfish. We are committed to providing a reliable and healthy source of farm-raised salmon to Canadians to help navigate future uncertainty to communities and boosting food security throughout Canada.

Performance against relevant actions and targets are summarized below.

ELIMINATE WASTE BY UTILIZING 100% OF THE PRODUCT TO SUPPORT A CIRCULAR ECONOMY

A circular economy retains and recovers as much value as possible from resources to reduce or eliminate waste. In the food system, this involves reducing food spoilage and waste through utilization of by-products and food waste and nutrient recycling. One example is fish by-products being used in pet food and fertilizers.

OUR PERFORMANCE

As an industry, we look at opportunities to utilize all parts of the salmon, even those that would not be fit for human consumption, such as trimmings. On average, 96.13% of the salmon is being utilized.



13.82% Animal Food

0.35% Fertilizer

2.69% Other

[6] *4 companies reporting (of 5 companies).*

SUPPORT ORGANIZATIONS THAT PROVIDE FOOD SECURITY TO CANADIANS

According to Statistics Canada, nearly 1 in 4 Canadians polled said they experienced food insecurity in their household because of financial constraints.

OUR PERFORMANCE

CAIA salmon farmers donated nearly 12,000 kg of salmon to food banks and other non-profit organizations that redistribute food to Canadians in need. According to the Canada Food Guide definition of one serving of salmon (1 meal = 75 g of salmon), this equates to 153,093 meals donated. In addition, CAIA salmon farmers donated $1,218,920.00 directly to non-profit organizations and First Nations that support food security amongst Canadians.[6]

CASE STUDY

Food security has become a rising social problem within the Mount Waddington region of British Columbia. Mowi Canada West supplies the Salvation Army with fresh Atlantic salmon for the people they serve through their numerous food distribution and food hamper programs. The program supports food security for houseless individuals, individuals dealing with mental illness and drug and alcohol addiction, and individuals escaping abuse, amongst others.





FOOD TRACEABILITY

COMMITMENT

Canadian farm-raised salmon is fully traceable from egg to plate. We understand the importance of having systems in place to make it easier for consumers to know where their food comes from and to provide greater assurance around our sustainability credentials. We will work to ensure consumers can more easily access the information they want about their salmon meals. Performance against relevant actions and targets are summarized below.



SUPPORT THE DEVELOPMENT OF CONSUMER-FRIENDLY TRACEABILITY SYSTEMS FOR CANADIAN FARM-RAISED SALMON

Food traceability is the ability to follow the movement of a food product and its ingredients through all steps in the supply chain.

OUR PERFORMANCE

CAIA salmon farmers are making efforts to support the transparency and traceability of their product. Internal company processes, such as tracking harvest lot and batch numbers, using population identification systems, and maintaining spawning records gives salmon farmers the ability to trace a product from the parental broodstock to the primary purchaser. While 100% of Canadian farm-raised salmon is traceable to the primary purchaser, more work is needed to make this traceability accessible to the consumer.

CONTINUE TO WORK TO RAISE THE LEVEL OF TRACEABILITY AWARENESS OF THE INGREDIENTS USED IN FISH FEED BY WORKING CLOSELY WITH OUR FEED SUPPLIERS

Feed traceability is the ability to track the movement of single ingredient feeds and mixed feeds. This is a regulatory requirement of the Canadian Food Inspection Agency. Feed companies, international standards bodies, and benchmarking organizations have been introducing enhanced systems to increase transparency in this area.

OUR PERFORMANCE

CAIA salmon farmers are working with feed suppliers to increase the traceability of fish feed. Companies are working to develop the right performance measure for this commitment.



OCEAN HEALTH

COMMITMENT

Healthy oceans are essential to healthy salmon farming. We are committed to continuous improvements to farm practices that further protect the marine ecosystem, including the health of the ocean floor (benthic) ecosystems beneath our farms, and to the prevention of plastic waste and debris from entering the oceans from our operations. Performance against relevant actions and targets are summarized below.

MONITOR FISH FEEDING BY USING UNDERWATER CAMERAS OR EQUIVALENT

Feed optimization and maximizing feed conversion ratios are of critical importance on farms.

OUR PERFORMANCE

Camera sensor units have software that enables real-time analysis of fish biomass and welfare to support feed optimization. 99.3% of ocean farms are equipped with underwater feed system monitoring equipment to minimize the impacts of feeding on the environment.

CASE STUDY

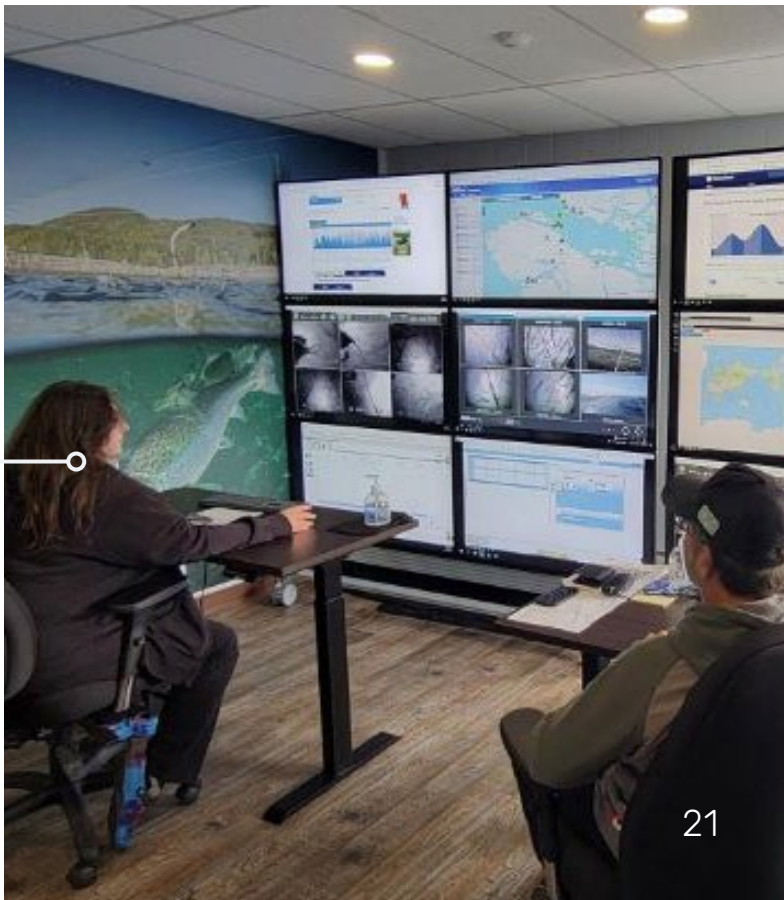
Mowi Canada West's installation of new centralized feeding stations in the villages of Klemtu and Port Hardy use state-of-the-art technology to support the company's local farms. This technology includes artificial intelligence that tracks individual feed pellets and fish activity levels to improve biological performance and to further reduce feed wastage.

WORK WITH PARTNERS TO INVESTIGATE AND ADOPT NEW TECHNOLOGIES AND INNOVATIVE PRACTICES TO MANAGE WASTE PRODUCED BY FISH

In all aquaculture operations, fish feces (organic waste) is released into the surrounding waters and could accumulate on the sea - or lakebed. Science-based government regulations are in place to ensure the impact of organic waste is well managed to ensure the ecosystems remain healthy.

OUR PERFORMANCE

In 2022, in accordance with local regulations, 100% of farm sites are remediated before a new generation of fish is transferred in. CAIA salmon farmers have implemented a number of practices to manage waste produced by fish, such as training of staff to reduce waste, and the use of automatic and centralized feeders.



SUPPORT RESTORATIVE AQUACULTURE, MULTI-SPECIES, AND/OR CO-PRODUCTION OPPORTUNITIES AT APPLICABLE LOCATIONS

In a report released in November 2021, an expert working group led by the Nature Conservancy defined restorative aquaculture as follows: "[R]estorative aquaculture occurs when commercial or subsistence aquaculture provides direct ecological benefits to the environment, with the potential to generate net positive environmental outcomes." The potential for restorative, multi-trophic, and or co-production operations is immense in Canada.

OUR PERFORMANCE

CAIA salmon farmers have made efforts to pursue restorative aquaculture, multi-species aquaculture, and/or co-production.[7] We have worked with First Nations, researchers and Fisheries and Oceans Canada (DFO) to investigate the feasibility of co-culture, such as shellfish and kelp. However, there are barriers to scaling these and other potential initiatives, including regulations for new products, regulatory uncertainty, and timelines. We are working to develop the right performance measure for this commitment.

ENSURE THAT NO FARMING EQUIPMENT IS RELEASED FROM FARM SITES, AND IN CASE OF ACCIDENT, THAT THERE IS PROMPT RECOVERY

The United States' National Office Oceanic and Atmospheric Administration provides the following description of marine debris: "Marine debris is defined as any persistent solid material that is manufactured or processed and directly or indirectly, intentionally or unintentionally, disposed of or abandoned into the marine environment or the Great Lakes". For the salmon farming sector, this would relate to equipment such as nets, tools, floats, pipes, etc. that can, in the case of accident, be released into the marine environment.

OUR PERFORMANCE

Companies have standard operating procedures on waste management to ensure that farm equipment is managed properly and does not result in harm to the local environment. For example, farming equipment is numbered and catalogued, and farms collect any equipment that is accidentally released from boats. Companies are working to develop the right performance measure for this commitment.

WORK WITH GOVERNMENTS, SUPPLIERS, AND INTERNATIONAL PARTNERS TOWARDS USING 100% REUSABLE, RECYCLABLE, OR BIODEGRADABLE PACKAGING

Polystyrene products (commonly known by the trademarked brand Styrofoam), are a common form of packaging in the food industry, including for salmon. Polystyrene is light weight, low cost, water resistant, and a good insulator, which is ideal for packaging seafood, which needs to be kept cool and sanitary. However, there are negative environmental impacts associated with polystyrene and there is a move away from the product.

OUR PERFORMANCE

Like many organizations that ship products around the world, CAIA salmon farmers are looking for opportunities to improve the use of reusable, recyclable, and biodegradable packaging materials at all stages of processing. Based on primary processing, where we have direct control, 86.47% of packaging used across the industry is reusable, recyclable, or biodegradable. However, it can be complex to understand fully how a product moves through the supply chain. We are working with governments, suppliers, and partners on improved packaging, including alternatives to Styrofoam packaging.

CASE STUDY

Aqua Pak is an EPS manufacturer providing packaging solutions for the aquaculture, pharmaceutical, and silviculture industries. For over 30 years, they have been helping the Canadian aquaculture industry increase their sustainability through product development and life-cycle innovations. Each year, Aqua-Pak recycles over 170,000 kg of consumer-sold materials for repurposing into reusable products, of which over 80% comes from aquaculture companies. The process is done in-house where the EPS is thermally re-densified into hardened ingots, which allows them to be transformed it into new forms of plastics such as park benches, picture frames, modular decking, and more. To support the reuse of EPS within the aquaculture space, Aqua-Pak accepts returns from their customers free of charge.

[7] *4 companies reporting (of 5 companies).*
[8] *3 companies reporting (of 5 companies).*

PUBLIC REPORTING

COMMITMENT

Transparency builds trust, strengthens accountability, and improves performance. We are committed to working with government bodies and other key stakeholders and rightsholders to provide information about our salmon farming operations to Canadians in a way that is easily accessible and understandable.

REPORT AGAINST THESE INDUSTRY-WIDE COMMITMENTS AT AN AGGREGATE LEVEL ON AN ANNUAL BASIS

The Canadian farm-raised salmon sector has extensive public reporting in place, via company-level annual reports, third-party certifications, and federal and provincial regulations. The industry is working together to aggregate performance to create additional transparency on progress. By bringing this publicly available information together in one report, we are making it easier for everyone to understand our commitment to sustainability.

OUR PERFORMANCE

CAIA salmon farmers intend to produce an annual sustainability report to measure performance against national commitments at an aggregate level. Ultimately, these reports will provide year-over-year performance trends. This document, published in 2024, is our first report.






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